

Addis Ababa University

College of Natural and Computational Sciences

Department of Computer Science



Final Year Project

Guideline

BSc in Computer Science

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1. Introduction

1.1 Purpose

The Final Year Project (FYP) is the capstone of the BSc Computer Science program. Students use it to bring together the knowledge and skills they have gained across the curriculum. Through the project a student must show problem-solving ability, technical competence, systematic development or research, evidence-based evaluation, professional documentation, and effective communication. This guideline sets out the technical, academic, supervision, assessment and defense framework for the project.

1.2 Authority and applicability

This guideline applies to every BSc-level Final Year Project of the Department of Computer Science, College of Natural and Computational Sciences, Addis Ababa University. It works within the University's senate legislation. Where this guideline and University regulations conflict, the University regulations prevail. The templates in the appendices are mandatory unless the FYP Committee approves an alternative in writing.

1.3 Core principle: the evidence chain

A project is not assessed solely by whether an application runs. The expected chain is:

Problem → Literature and gap → Objectives → Methodology → Requirements → Design → Implementation → Testing → Evaluation → Conclusion.

Documented evidence must connect each stage to the next. A break in the chain counts as a weakness even when the software works.

1.4 Learning outcomes

By the end of Final Year Project I and Final Year Project II, a student should be able to:

- identify and analyze a relevant problem;
- formulate measurable objectives;
- review and synthesize related work critically;
- select and justify methods and technologies;

- design and implement a solution;
- test and evaluate the solution against the objectives;
- plan and manage a project, including its risks;
- work responsibly in a team and apply professional, ethical and legal standards;
- document all technical work; and
- present and defend the documents and the implementation.

1.5 Terminology and conventions

In this guideline, “must” marks a mandatory requirement, “should” marks a strong recommendation that needs a stated reason if it is not followed, and “may” marks an option. Table 1.1 defines the main terms.

Table 1.1: Key terms used in this guideline

Term	Meaning
Final Year Project I (FYP I)	The first course in the sequence. It covers title selection, proposal, literature review, requirements, analysis and design.
Final Year Project II (FYP II)	The second course. It covers implementation, testing, evaluation, the final report, and the final defense.
Coordinator	The staff member appointed to run the FYP I and II courses day to day. The Coordinator chairs the Final Year Project Committee or acts for it.
Final Year Project Committee	The committee appointed by the Department Head to administer the project courses. The Coordinator chairs it or acts for it.
Group	The students who carry out one project together. A group may also be a single student where Section 14.1 allows it.
Advisor	The academic staff member who supervises the group.
Co-advisor	A second supervisor, internal or external, appointed for specialist expertise.
Examiner	A staff member, other than the advisor, who assesses the submitted evidence and conducts the defense.
SO1, SO2, ...	The numbered specific objectives of a project (Section 5.2).
Deliverable	Any item the group must submit, such as a proposal, a report, a system or a logbook.

2. Project Structure, Types and Complexity

2.1 Structure of the capstone sequence

The project runs over two semesters. Unless the curriculum states otherwise, Final Year Project I is a prerequisite for Final Year Project II. Table 2.1 shows what each course covers and what is assessed.

Table 2.1: Capstone course structure

Course	Main focus	Principal deliverables	Closing gate
Final Year Project I	Title selection, proposal, literature review, requirements, analysis and design	Proposal, FYP I document (Chapters 1 to 4 of the report), presentation, logbook	Final Year Project I defense
Final Year Project II	Implementation, testing, evaluation, final report	Working system or research outputs, final report, test and evaluation evidence, presentation, demonstration, logbook	Final Year Project II defense

2.2 Guiding principles

Projects should be relevant, feasible, technically challenging, evidence-based, ethically responsible, and evaluated against explicit objectives. Every member must be able to defend the work, individually and as part of the group. Assessment rewards demonstrated competence and not presentation polish alone.

2.3 Acceptable project categories

Acceptable categories are software and application development; artificial intelligence and machine learning; data science and analytics; networking and distributed systems; cybersecurity; software engineering research; algorithms and theoretical computing; simulation and modeling; Internet of Things (IoT) and embedded systems; and interdisciplinary or hybrid projects. A project may combine categories, but the

proposal must name one primary category so that the right assessment emphasis is applied.

2.4 Complexity expectation

A BSc project should involve several substantial activities. A project of standard complexity includes at least five of the following: requirements analysis, architecture and design, database or data engineering, algorithm selection or model development, substantial programming, systematic testing, quantitative or user evaluation, security design, deployment, and practical research. A simple CRUD application with minimal analysis, design, testing and evaluation is classified as low complexity. Table 2.2 describes the three classes.

Table 2.2: Complexity classes

Class	Description	Treatment
Low	Fewer than five activities, mostly standard forms and reports, little analysis or evaluation, no meaningful algorithmic or data content.	Not approved. The group must extend the scope before approval.
Standard	At least five of the activities listed above, with a clear technical challenge and measurable evaluation.	Approved. Eligible for any mark band.
High	Standard complexity plus a substantial research, data, algorithmic, security or systems challenge, or a real external stakeholder with real data.	Approved. Required if a fourth group member is requested.

The Committee judges the complexity of the project at proposal approval, and the examiners confirm it at the defense. If examiners find that an approved project is in practice of low complexity, they must score Technical Implementation no higher than the “Adequate” band in Section 17.6.

3. Roles and Responsibilities

3.1 Students

Students must:

- select and understand the problem;
- prepare deliverables on time;
- conduct the work independently;
- maintain records, including the logbook and the version-control repository;
- implement and test the solution;
- prepare all required documentation and evidence;
- disclose limitations;
- protect data;
- declare any use of artificial intelligence (AI) tools and third-party material; and
- defend the work.

3.2 Advisor

The advisor must:

- approve scope and milestones;
- provide academic and technical guidance;
- review progress, including the repository and logbook;
- challenge weak assumptions;
- verify that progress is substantial;
- advise on ethics and risk;
- complete the progress assessment in Appendix C; and
- recommend whether the group is ready for defense.

The advisor guides the work and does not implement it.

3.3 Co-advisor and external supervisor

The Committee may appoint a co-advisor when the topic needs expertise that the advisor lacks, for example from industry, another department or a research institute. An external supervisor must confirm the problem, the data access and the sponsor's expectations in writing. The internal advisor stays accountable for supervision and for the progress assessment. A co-advisor or external supervisor may give feedback to the examiners but does not score.

3.4 Final Year Project Coordinator and Committee

The Coordinator or Committee is responsible for:

- coordinating title approval and ethical screening;
- allocating advisors;
- preparing submission and presentation schedules;
- preparing common templates;
- monitoring milestones;
- assigning examiners;
- arranging defense logistics;
- running similarity checks and managing the archive;
- publishing the maximum number of groups per advisor each semester (a maximum of five is recommended); and
- consolidating grades.

3.5 Examiners

Examiners must:

- assess the submitted evidence independently;
- evaluate technical and academic quality;
- verify each member's individual understanding;
- conduct the defense;

- record strengths and deficiencies on the assessment sheet (Appendix N); and
- recommend a final mark according to the approved rubric.

Each defense has at least one examiner, who must not be the advisor of the group. Where the Committee appoints more than one examiner, none of them may be the advisor of the group. An examiner must declare any conflict of interest, such as a family or financial relationship or earlier co-supervision of the work, and must step aside from the evaluation of that group. The Coordinator provides each examiner with the report at least seven days before the defense.

4. Title Selection, Proposal and Approval

4.1 Sources of project ideas

Project ideas may come from academic staff, from industry, government or community partners, from research projects in the Department, or from the students themselves. Projects that address a real problem for identifiable stakeholders are preferred, including problems in the local context and national priorities. A sponsored project needs a letter from the sponsor that names the problem, the contact person, and any data the sponsor will provide.

4.2 Selection criteria

A title should be judged on the following criteria:

- relevance to the discipline;
- significance of the problem;
- feasibility within the academic period;
- availability of data and resources;
- technical depth;
- availability of appropriate supervision;
- ethical acceptability;
- potential contribution;
- manageable scope; and
- originality, meaning that the project is not a repeat of an archived project unless it extends that project substantially.

4.3 Proposal requirements

The proposal must contain the following sections. Appendix A gives the template.

- cover page;
- background;
- problem statement and gap;

- motivation;
- general objective;
- specific objectives;
- research or development questions, where applicable;
- scope and limitations;
- significance;
- literature review and related work (a focused review is required for every project);
- methodology;
- preliminary requirements;
- proposed design and technologies;
- resources, risks and ethical considerations;
- expected deliverables;
- work plan; and
- references, with appendices where appropriate.

4.4 Proposal quality test

A reviewer should be able to answer these questions from the proposal alone:

- What problem exists?
- Why does it matter?
- What gap is being addressed?
- What exactly will be produced or investigated?
- How will success be measured?
- Is the plan feasible with the time, data, tools and skills available?
- Is the project ethical and legally permissible?

4.5 Approval gate

Development must not start until the Committee has approved the project title, problem statement, objectives, scope, methodology, expected deliverables, resources, risks and advisor. The advisor recommends and the Committee decides. The decision is recorded in the approval block of Appendix A. Ethical screening under Appendix M is part of the gate.

4.6 Change control

A change is major if it alters the problem, the general objective, any specific objective, the primary dataset or technology, the group membership, or the advisor, or if it delays a milestone by more than two weeks. A major change needs a written request that gives the reason and the effect on scope, schedule and testing. The advisor endorses the request and the Committee approves it before implementation expands. The Committee keeps the record. The project title may change only before the Final Year Project I defense. Minor wording changes may be made at any time with the advisor's consent.

5. Problem Statement, Objectives and Literature Review

5.1 Problem statement

The problem statement must describe the existing condition, the affected stakeholders, evidence that the problem exists, its consequences, the shortcomings of current approaches, and the specific gap that the project will address. Evidence should come from the group's own investigation, such as interviews, observation, document analysis or data, or from credible sources. Avoid generic statements such as "the existing system is manual; therefore, a computerized system will be developed."

5.2 Objectives

State the general objective and the specific objectives of the project. The general objective describes the ultimate goal of the project and is written as a single sentence. The specific objectives are the steps that together achieve the general objective, and each one must contribute to it. Specific objectives must be measurable, technically relevant, realistic in terms of the capacity of the group, the available resources and the time, and traceable to requirements and to evaluation evidence. A project normally has four to seven specific objectives, numbered SO1, SO2 and so on, so that they can be referenced in the traceability matrix (Appendix E). Begin each one with an action verb such as analyze, design, implement, compare, evaluate, validate or optimize, and avoid vague verbs such as know, understand or explore. Do not list general or obvious activities as specific objectives. For example, drawing conclusions must not be written as a specific objective, because every project ends with conclusions whether or not they are stated as an objective. The same applies to writing the project report. The objectives may be modified during the work, but any change to the general objective or to a specific objective is a major change and follows Section 4.6. The headings General Objective and Specific Objectives carry no section numbers and are not listed in the table of contents.

A weak objective reads: “To develop a good system for the clinic.” A stronger one reads: “To implement a triage-priority classifier for outpatient records that reaches at least 85% recall on the high-priority class on a held-out test set.”

5.3 Research and development questions

Where applicable, state the questions the project will answer. Each question must link to an objective and have a defined way of being answered. Experiment-based projects should state their hypotheses.

5.4 Literature review standard

A literature review is a critical synthesis of existing knowledge. It is not a summary of the literature and it is not a series of separate descriptions of individual studies. Its purpose is to show what is already known about the problem, how others have addressed it, where their findings agree or conflict, and what remains unresolved, so that the proposed project is justified by evidence.

Students must review credible and relevant sources, including recent work alongside foundational work. The review must be organized by theme, problem, approach or technology, and not source by source. Within each theme the student must:

- group the sources that address the same question or use similar methods, and state what they have in common;
- compare and contrast the approaches in terms of methods, technologies, data, evaluation measures, results, strengths and weaknesses;
- judge the quality of the evidence critically, for example by noting small or unrepresentative data, weak evaluation, outdated technology, or results that cannot be compared;
- identify agreement, disagreement and open questions across the sources; and
- state, in the words of the student, what the evidence shows as a whole.

The review must end with an explicit gap statement. It must say what existing work does not do, why this matters for the stakeholders named in the problem statement, and how the proposed project responds to the gap. The review must also show how the evidence shaped the choice of methods, technologies and design. Sources that do not support the argument of the review should be left out, because the number of sources is not a measure of quality.

Students must express the ideas of others in their own words and cite them (Section 11.11). A source that the student has not read must not be cited, and a summary produced by an AI tool must not replace reading the source (Section 15.3). Copying or closely paraphrasing the text of a source is treated under Section 15.1.

5.5 Minimum evidence

The review should include recent sources alongside foundational work. It should distinguish related systems or studies from the proposed work and show how the reviewed evidence informed design or methodology decisions. A proposal should normally cite at least 15 credible sources, and the final report should extend that list. A comparison table must compare at least three directly comparable systems or studies on the same stated criteria, for example the problem addressed, method or technology, data, evaluation measures, results, strengths and limitations. The table supports the synthesis and does not replace it. Section 11.11 sets the standard for source quality.

6. Methodology and Project Planning

6.1 Method selection

The group must justify the research or software-development methodology it selects. Acceptable approaches include Agile or Scrum, Waterfall, iterative development, prototyping, Spiral, design science, and experimental research where appropriate. The methodology must show how each specific objective will be achieved and how success will be measured.

6.2 Project plan

The plan must identify tasks, dependencies, milestones, responsible persons, expected outputs, risks and mitigation actions. A work breakdown structure or Gantt chart is recommended. The milestones must align with the timeline in Section 16.2. The group must update the plan when major changes occur.

6.3 Risk management

The group must keep a risk register (Appendix H) and review it with the advisor at least once a month. Typical risks are unavailable data, technology failure, scope expansion, schedule delay, and security or privacy problems. For each high risk the plan should name a fallback, such as a substitute dataset or a reduced minimum scope.

6.4 Feasibility and resources

The proposal must state the hardware, software, data, compute and budget the project needs, and who will provide them. The Department does not guarantee resources beyond its standard laboratories. Open-source tools are preferred. Where the target setting has limited connectivity or power, the design should say how the system behaves in those conditions.

7. Requirements, Analysis and Design

7.1 Functional requirements

Document what the system must do in clear, testable statements. Give each requirement a unique identifier (for example FR-01) and a priority. Requirements must be traceable to objectives and to acceptance tests.

7.2 Non-functional requirements

Consider measurable quality attributes such as performance, security, usability, reliability, maintainability, scalability, availability, accessibility and compatibility, and add localization and offline operation where they apply. State each attribute with a measurable target. For example: "The search page returns results within two seconds for 95% of requests with 50 concurrent users."

7.3 Requirements baseline

A requirements specification must be approved by the advisor at the requirements baseline gate in Table 16.1, which falls in weeks 5 to 7 of Final Year Project I and before the design phase begins. The Committee confirms it at the Final Year Project I defense. Record each later change with its rationale and its effect on scope, schedule and testing.

7.4 Recommended artifacts

Depending on the project, use a use-case diagram, activity diagram, sequence diagram, class diagram, state diagram, relational schema, data dictionary, architecture diagram, component or deployment diagram, data-flow diagram, wireframes, algorithm or pseudocode, and a prototype. Use UML 2.x or an equivalent recognized notation and explain any other notation.

7.5 Consistency rule

Diagrams must represent the actual proposed or implemented system. Requirements, models, database structures, interfaces and implementation must not contradict one another. Update the design documents when the implementation changes.

8. Technical Implementation

8.1 Software quality

The implementation should show appropriate architecture, modularity, readability, maintainability, error handling and input validation, suitable algorithms and data structures, and reasonable technology selection. The code should follow a stated coding standard, use meaningful names, and carry comments where the logic is not obvious. The repository must include a README that explains how to build and run the system.

8.2 Database quality

Where applicable, document the schema design, relationships, constraints, normalization, indexing, transactions, security, and backup and recovery considerations. Provide creation and migration scripts and seed data that contains no real personal data.

8.3 Security

Where relevant, address authentication, authorization, password protection, input validation, secure sessions, access control, secure communication and common application vulnerabilities, using a recognized reference such as the Open Worldwide Application Security Project (OWASP) Top 10. Consider security throughout development. Passwords, application programming interface (API) keys and other secrets must not be committed to the repository. Dependencies should be checked for known vulnerabilities.

8.4 Version control and collaboration

Groups must use a version-control system. The repository must be accessible to the advisor and examiners, and each member must commit their own work under their own identity. Meaningful commit messages, branches, pull requests, issue tracking and milestone tags are expected. This evidence shows progress and individual contribution, but examiners will not treat it as the only evidence of either.

8.5 Reproducibility, deployment and licensing

Another person must be able to build and run the system from the submitted materials. List dependencies with their versions, keep configuration separate from code, and use containers or scripted setup where practical. The system must be demonstrable in an environment available at the defense. List all third-party libraries, data and media with their licenses, and use them only within those licenses.

8.6 Technical documentation

The submission must include the technical documentation described in Appendix J, together with an installation guide and a user guide.

9. Artificial Intelligence, Machine Learning and Data-Driven Projects

This section applies in addition to Sections 5 to 8 and Section 10 to every project that trains, tunes, evaluates or deploys a machine-learning model, or that draws conclusions from data.

9.1 Problem framing and baselines

The proposal must define the task, the inputs and outputs, and the success criteria before modeling starts. It must explain why machine learning is needed instead of a simpler rule-based solution. Every project must report at least one baseline, such as a majority-class predictor, a rule-based method, a simple linear model, or a previously published system, evaluated under the same conditions as the proposed model.

9.2 Dataset

Document the source, size, variables or features, target, collection method, licensing and permissions, preprocessing, and ethical considerations. For labeled data, also describe how labels were produced, who produced them, and how agreement between annotators was measured. Report the class distribution and describe how the train, validation and test sets were created. Personal data must be anonymized or pseudonymized.

9.3 Modeling

Explain the baseline selection, algorithm choice, feature engineering, train, validation and test strategy, hyperparameters, training procedure, and reproducibility measures such as random seeds, library versions and environment files. Split the data before any preprocessing that learns from it, and never tune on the test set. With small datasets, use cross-validation. State the compute used and the training time.

9.4 Evaluation

Select metrics appropriate to the problem. Classification may use precision, recall, F1, ROC-AUC and confusion matrices. Regression may use MAE, MSE, RMSE or R^2. Other problems need suitable domain-specific metrics. Report the mean and the variation across runs or folds, and use a significance test when claiming that one method is better than another. Include an error analysis that examines representative failures. Discuss class imbalance, overfitting, generalization, bias, limitations and interpretability where relevant.

9.5 Projects that use pretrained or generative models

A project that builds on a pretrained, foundation or generative model must state the model name, version and provider, its license and terms of use, and exactly what the group contributed, for example fine-tuning, retrieval, data preparation or evaluation. An application that only calls an external model through its interface does not by itself meet standard complexity. The group must evaluate outputs systematically on a defined test set, for example for accuracy, groundedness, hallucination rate and unsafe output, and must record the prompts, parameters, cost and latency. Sensitive or personal data must not be sent to an external service without prior approval.

9.6 Responsible AI

Where relevant, assess fairness across user groups, privacy risks, transparency, and potential misuse. Document the intended use, the known limits and the data used in a short model description (model card) and a data description (datasheet). Projects on Ethiopian languages should record the script, normalization choices and dialect coverage.

9.7 Reproducibility package

Submit the code, configuration, environment file, instructions for obtaining the data or a permitted sample, the trained model or the training script, and a table that links each reported result to the script that produced it.

10. Testing, Quality Assurance and Evaluation

10.1 Testing levels

Use appropriate unit, integration, system, acceptance, performance, usability and security testing. Not every project needs every level, but the test plan must justify each exclusion. Automated unit tests are expected for core logic where feasible. Security testing must stay within the limits in Section 15.6.

10.2 Test planning and evidence

Prepare a test plan during design. Each test case must contain an ID, the requirement it covers, the input or precondition, the expected result, the actual result, the status, and where useful, evidence such as screenshots or logs. Appendix F gives the template. "The system was tested and works" is not sufficient evidence.

10.3 Defect management

Record important failures, analyze them, correct them where possible, and re-test. Discuss known unresolved defects and their impact in the report.

10.4 User evaluation

Where the system has human users, evaluate it with representative users on defined tasks, using a recognized instrument such as the System Usability Scale or a task-success measure. Explain how many participants were used and why that number is adequate. Participant studies need informed consent and ethical screening under Section 15.4.

10.5 Objective-to-evidence principle

Every specific objective must have a corresponding implementation or research output and an evaluation method. The final report must state whether each objective was achieved, partially achieved, or not achieved, and must cite the evidence.

10.6 Traceability matrix

Use the matrix in Appendix E to connect objectives, requirements, design, implementation, tests and results. Start the matrix in Final Year Project I and complete it in Final Year Project II.

11. Project Report and Documentation Standard

11.1 Writing standard

Use clear technical English, consistent terminology, numbered headings, correctly labeled figures and tables, a consistent citation style, and evidence-based claims. Avoid excessive screenshots, copied diagrams and unsupported claims.

The Department of Computer Science sets the minimum format for the final project documents. The structure in Table 11.1 is mandatory. Within that structure, the group decides the detailed content and length with the advice of the advisor. Grammar, punctuation, spelling and other mechanical matters are the sole responsibility of the group members.

11.2 Report structure

The Final Year Project I document contains Chapters 1 to 4 of the structure in Table 11.1. The Final Year Project II report is the complete document, with Chapters 1 to 4 updated to match the final system and Chapters 5 to 7 added.

Table 11.1: Required structure of the project report

Part	Contents
Front matter	Title page (Appendix K); declaration; approval sheet; acknowledgements (optional); abstract of 200 to 300 words with keywords; table of contents; list of tables; list of figures; list of abbreviations.
Chapter 1: Introduction	Background, problem statement, motivation, general and specific objectives, research or development questions, scope and limitations, significance, organization of the report.
Chapter 2: Literature review and related work	Thematic synthesis of the literature, comparison table, gap statement.
Chapter 3: Methodology	Development or research method, tools, data, work plan, risk management.

Chapter 4: Requirements analysis and design	Requirements specification, models, architecture, database or model design.
Chapter 5: Implementation	Main components and algorithms, key code excerpts, deployment.
Chapter 6: Testing, results and discussion	Test evidence, evaluation against each objective, traceability matrix, limitations.
Chapter 7: Conclusion and recommendations	Summary against the objectives, contributions, limitations, future work.
Back matter	References; appendices such as the user guide, extra diagrams, test cases, dataset documentation, AI-use declaration and contribution statement.

11.3 Page and font specification

Follow the specifications below throughout the document. For the title pages, use the template in Appendix K.

11.3.1 Paper size

Use standard A4 paper (8.27 in by 11.69 in, or 210 mm by 297 mm).

11.3.2 Font

Table 11.2: Font specification

Element	Font	Size (pt.)	Style
Body text	Times New Roman	12	Regular
Major heading (chapter title)	Times New Roman	14	Bold, upper case, centered
Second-order heading	Times New Roman	13	Bold
Third-order heading	Times New Roman	12	Bold
Captions	Times New Roman	11	Regular, label in bold
Footnotes	Times New Roman	10	Regular
Code listings	Courier New	10	Regular, single spaced

Font color must be black unless another color carries a specific meaning. Headings deeper than the third order should be avoided.

11.4 Spacing and paragraphs

Use 1.5-line spacing throughout, and justify the alignment of each paragraph. Lists in the main text also follow the 1.5 rule. Single spacing is used for captions, tables, figure contents, code listings, footnotes, the table of contents and lists of tables and figures, entries of the reference list, and the forms and tables in the appendices. Text inside tables, captions, headings and reference entries may be left aligned.

Set 6 points of spacing after each paragraph. New paragraphs start on the next line, with no blank line between paragraphs and no first-line indent. Each chapter must start on a new page. Chapter titles are centered Major Headings.

11.5 Margins

Use the margins in Table 11.3 for the whole document. The wider left margin allows for binding.

Table 11.3: Document margins

Top	Bottom	Left	Right
1.00 in (2.54 cm)	1.00 in (2.54 cm)	1.25 in (3.18 cm)	1.00 in (2.54 cm)

11.6 Tables

Number tables consecutively within each chapter, using the chapter number first. For example, Table 3.1 is the first table of Chapter 3. Write the label and caption above the table, refer to every table in the text, and give units in the column headings. Do not paste tables as screenshots.

11.7 Figures

Number figures in the same way, for example Figure 3.1, and write the caption below the figure. Figures must be legible when printed. Draw diagrams yourself or credit the source at the end of the caption.

11.8 Equations and code listings

Number equations in parentheses at the right margin, for example (3.1), and define every symbol. Show only essential code excerpts in the report body. Full code belongs in the repository and, where needed, in an appendix.

11.9 Page numbers

Do not number the title page. Number all pages that come before the first page of Chapter 1 with lower-case Roman numerals (i, ii, iii, iv, ...). Number the pages from the first page of Chapter 1 onward with Arabic numerals (1, 2, 3, ...). Align page numbers to the right.

11.10 Headers

The header carries the title of the project report on odd pages and the title of the chapter or section on even pages. The first page of every chapter or section has no header.

11.11 References and citation

The Department requires the IEEE reference format. Cite references in the text and in the captions of figures and tables with numbers in square brackets, for example [1] or [2], [5]. Number the references with Arabic numerals in the order of first citation, and list them in that same order at the end of the document under the heading References. Do not cite references in chapter, section or subsection titles. A citation tells the reader that an idea, concept, formulation, data item, inference or piece of information is attributable to the cited work. Every figure or table taken from the literature must be acknowledged at the end of its caption with the reference number.

The main reference sources are books, monographs, handbooks, archived journal papers, conference papers in published proceedings, institutional technical reports, theses and project documents, dissertations, other archived reports, and standards. Internet sources are increasingly important, but Internet references must not form the entire list. Web material may be transitory, may not be technically reviewed,

and may have doubtful authenticity. Use it as a secondary source to supplement the main sources. Check that every cited work exists and says what you attribute to it.

Generative AI tools must not be listed as authors. Disclose their use under Section 15.3. Cite an AI tool only when you quote or reproduce its output.

The patterns and examples below show the expected IEEE format. Write author names with initials first, give article and chapter titles in sentence case inside quotation marks, abbreviate journal names as in the IEEE style, and write every access date in the form (accessed Mon. DD, YYYY).

For a book: A. A. Author, *Title of Book*, xth ed. City of Publisher, State or Country: Publisher, year, ch. x, pp. xxx–xxx.
For a journal paper: A. A. Author, "Title of article," *Abbrev. Title of Journal*, vol. x, no. x, pp. xxx–xxx, year, doi: xxxxx.
For a proceedings paper: A. A. Author, "Title of paper," in *Proc. Name of Conference*, City, Country, year, pp. xxx–xxx.
For an Internet source: A. A. Author or Organization, "Title of page," Website Name. [Online]. Available: URL (accessed Mon. DD, YYYY).
For software or a dataset: A. A. Author or Organization, "Title," version x.x, year. [Online]. Available: URL or DOI (accessed Mon. DD, YYYY).

Examples:

[1] L. R. Rabiner and B. H. Juang, *Fundamentals of Speech Recognition*. Englewood Cliffs, NJ, USA: Prentice Hall, 1993.

[2] **ብላታ መርስዔ ኀዘን ወልደ ቂርቆስ፤ ያማርኛ ሰዋሰው:: አዲስ አበባ፣ አርቲስቲክ ማተሚያ ቤት፣ 1948 ዓ.ም::**

[3] S. T. Abate and W. Menzel, "Syllable-based speech recognition for Amharic," in *Proc. 2007 Workshop Comput. Approaches Semitic Lang.: Common Issues and Resources*, Prague, Czech Republic, 2007, pp. 33–40.

[4] S. H. Asefa and Y. Assabie, "Transformer-based Amharic-to-English machine translation with character embedding and combined regularization techniques," *IEEE Access*, vol. 13, pp. 1090–1105, 2025, doi: 10.1109/ACCESS.2024.3521985.

[5] M. P. Lewis, Ed., *Ethnologue: Languages of the World*, 16th ed. Dallas, TX, USA: SIL International, 2009. [Online]. Available: http://www.ethnologue.com/ (accessed Oct. 17, 2026).

[6] scikit-learn developers, “scikit-learn: Machine learning in Python,” version 1.5, 2024. [Online]. Available: https://scikit-learn.org (accessed Oct. 17, 2026).

11.12 File format and submission copy

Submit the final report as a searchable PDF file together with its editable source (DOCX or LaTeX). Name the file with the group identifier, the course and the year. The Department decides whether bound hard copies are also required.

12. Presentation Slide Formatting

12.1 Purpose and structure

Each group defends its project with a presentation. Prepare an outline first, so that the slides give a clear message: the problem, the objectives, the method, the design, the main implementation points, the evaluation results, the limitations and the future work.

12.2 Slide layout

- Use a white background with little or no graphics.
- Keep bullet points short, written as phrases and not sentences, and use no more than about six bullets per slide.
- Leave white space around the main content so that attention stays on the key information.
- Make every figure and table legible from the back of the room.

12.3 Font

- Use the Arial font in black.
- Set the main body text at 20 to 24 points.
- Use the same font, size and style for the main content area on all slides except the title slide.
- Set the project title on the title slide at 36 points at most.
- Give each content slide a brief, descriptive title that reflects its content and is not a full sentence. Set slide titles at 30 points.

12.4 Number of slides and timing

A general rule is one slide per minute. A 20-minute presentation should have about 20 slides, and no more than 22 slides, including the title slide, are allowed for the defense. Backup slides placed after the final slide are not counted. Focus on the

key concepts the audience should remember. Submit the slides to the Coordinator at least 48 hours before the defense.

12.5 Common mistakes

Slides with too much text distract the audience from the speaker. Text-only slides should contain short highlights written as phrases.

13. Advising, Progress Monitoring and Logbook

13.1 Meetings

The group must meet the advisor at least once every two weeks, on a schedule they arrange. Each meeting needs a short agenda and documented outcomes, and the advisor signs or initials the logbook entry.

13.2 Logbook

The group keeps a logbook (Appendix B) and updates it every week. Each entry records the date, the objective addressed, progress, issues, advisor feedback, agreed actions and the next milestone, and the advisor initials the entries after each meeting. The logbook must show which member did what. It supports supervision and the assessment of individual contribution.

13.3 Progress review and readiness for defense

Before each defense the advisor completes the progress assessment in Appendix C for every member and gives one of three recommendations: ready for defense, ready with minor corrections, or not ready. If the advisor recommends "not ready", the Committee decides whether the group defends in the current session or in a later one. The advisor must give the group the reasons in writing.

13.4 Change of advisor or advisor absence

If an advisor becomes unavailable for more than two weeks, or a serious breakdown in the working relationship occurs, the Coordinator assigns a replacement within one week. The replacement receives the logbook and the repository.

14. Group Projects and Individual Contribution

14.1 Group size

A group normally has three students, and the maximum is four. The Department may announce a different standard size. A group of fewer than three students, including an individual project, is allowed only with the written approval of the Committee, and the scope of the project is then adjusted to the size of the group. A fourth member is allowed only for a project of high complexity that the Committee approves.

14.2 Roles

Each member must hold primary technical responsibility for at least one major component or work package. The roles are declared in the proposal and recorded again in the Individual Contribution Statement (Appendix G).

14.3 Individual accountability

Each student must be able to explain the architecture and overall operation of the system, in addition to demonstrating their own contribution. Examiners should ask about individual modules, code, design decisions and troubleshooting. The examiners record the result as an Individual Contribution Factor, as described in Section 17.7.

14.4 Disputes and withdrawal

A dispute within a group goes first to the advisor for mediation and then to the Committee. A member who wishes to withdraw must give written notice. The remaining members then adjust the scope through change control (Section 4.6). A departing member may not use the group's work without the consent of the other members. If a member does not contribute despite documented warnings from the advisor, the Committee may remove that member from the group or assign an adjusted mark.

15. Academic Integrity, Ethics and Responsible Use of AI Tools

15.1 Integrity

Plagiarism, purchased projects, fabricated data or results, misrepresented authorship, unauthorized copying, and the undisclosed reuse of earlier students' projects are prohibited and are subject to university regulations.

15.2 Similarity checking

The Committee screens every report with a similarity-checking tool before the defense and compares source code with the archive and with public repositories where tools allow. An overall similarity above 20%, or more than 5% from a single source, with references and properly quoted text excluded, triggers a review by the advisor and the Committee. A high score is not proof of plagiarism, and a low score is not proof of originality. The review looks at the sources and at how the text is used.

15.3 Responsible use of AI tools

Where institutional policy permits AI assistance, students remain responsible for correctness, originality, security, citations, testing and disclosure. The following rules apply.

- AI tools may be used, with the advisor's knowledge, for brainstorming, language editing, code assistance, debugging and finding literature.
- Students must not present AI-generated results or data as real measurements, and must not cite sources they have not verified. Every reference must be checked.
- AI-generated code and text must be understood, reviewed and tested by the student. Substantial AI-generated code should be attributed in the code comments.
- Confidential or personal data must not be entered into public AI services.

- AI tools must not be used to answer questions during the defense.
- Each group must keep a record of significant AI use (tool, version, date, purpose) and submit the declaration in Appendix L with every report.

Examiners may test a student's understanding of any AI-assisted part. A student who cannot explain such work is treated under Section 18.1. A stricter University or Department policy overrides this section.

15.4 Data ethics and ethical clearance

Projects that involve personal, health, educational, financial, biometric or other sensitive data must follow the applicable institutional and legal requirements, including Ethiopian data protection law. These include consent, data minimization, access control, anonymization or pseudonymization where appropriate, secure storage, and deletion when the data is no longer needed. Every proposal must complete the ethical screening checklist in Appendix M. A project that involves human participants, identifiable personal data, health, biometric, financial or education records, minors or other vulnerable people, or confidential sponsor data needs the written approval of the advisor and the Committee, and must be referred to the college or University ethics review where that review is required. Collecting data from websites or platforms must respect their terms of use and the law.

15.5 Intellectual property, confidentiality and licensing

Subject to University policy and to any written sponsor agreement, students keep the copyright in their own work. A sponsored project needs a written agreement before work starts that covers ownership, confidentiality and publication. The Department may archive reports and use them for teaching and accreditation. Third-party code, data and media may be used only within their licenses and must be listed. A group that releases its code as open source should choose a license with the advisor.

15.6 Safety and authorized testing

Hardware and IoT work must follow laboratory safety rules. Security testing must be carried out only on systems that the group owns or has written permission to

test, and offensive security work must run in an isolated laboratory environment within an authorized scope. Groups must not release malicious software.

16. Milestones, Deliverables and Submission

16.1 Deliverables

Groups must submit the following deliverables according to the schedule prepared by the Committee or Coordinator:

- project proposal;
- Final Year Project I document and Final Year Project II report;
- presentations;
- implementation;
- demonstrations; and
- logbook.

Examiners consider the quality of content, format, English grammar and punctuation. Timeliness is governed by the late-submission rule in Section 16.3 and is not scored separately.

16.2 Recommended two-semester timeline

Table 16.1 gives the indicative timeline. The Department should map these weeks to its official academic calendar.

Table 16.1: Recommended two-semester timeline

Course	Phase	Weeks	Deliverable	Gate
FYP I	Orientation and title selection	1 to 2	Topic concept	Advisor or Coordinator approval
FYP I	Proposal	3 to 4	Proposal and presentation	Proposal approval
FYP I	Extended literature review and requirements analysis	5 to 7	Extended literature review and requirements specification	Requirements baseline
FYP I	Design	8 to 13	Models, architecture, database and UI design	Design review

FYP I	Final Year Project I defense	14	Final Year Project I document and presentation	Presentation and assessment
FYP II	Implementation I	1 to 4	Core prototype	Progress review
FYP II	Implementation II	5 to 8	Integrated system	Prototype review
FYP II	Testing and evaluation	9 to 10	Test results and evaluation	Evaluation review
FYP II	Final report	11 to 13	Draft final report	Advisor approval
FYP II	Final Year Project II defense	14	Report, presentation and demonstration	Final assessment
FYP II	Corrections and final submission	15 to 16	Corrected report and submission package	Advisor verification

Where the defense outcome requires major corrections, the correction period follows Section 17.9 and may extend beyond week 16.

16.3 Late submission

The Committee announces the late-submission penalty schedule at the start of each semester. Unless it announces otherwise, a deliverable loses 5% of its maximum mark for each day of delay, and a deliverable more than five days late is treated as not submitted. The Committee may waive the penalty for documented illness, bereavement or other circumstances beyond the group's control when the group reports them in writing before the deadline or within three days after it.

16.4 Submission package

The final submission package consists of the final report; source code; database and scripts where applicable; the executable or deployed system where applicable; the dataset where permitted; the installation guide; the user guide; testing evidence; the presentation slides; the logbook; the individual contribution statement; the AI-use

declaration; the similarity report; the reproducibility package for AI and data projects; and the required declarations and approvals. Appendix I is the checklist.

16.5 Archiving

The Committee archives the approved final report as a PDF, together with a snapshot of the repository. The archive is used to check originality in later years. Sensitive data and confidential sponsor material are excluded from the archive.

17. Defense and Assessment

17.1 Defense panel

The panel consists of a chair, who is the Coordinator or a delegate, and at least one examiner. The advisor attends, may clarify matters, and does not score the examiner component. All group members must attend. A member who is absent without an accepted reason does not defend and is treated under Section 18.1.

17.2 Defense procedure

The defense should include a concise problem and objective overview, the methodology, key design decisions, a live or recorded demonstration as approved, evaluation results, limitations and future work. Questions should test understanding and not memorization. Table 17.1 shows the default time allocation, which the Committee may change by announcement.

Table 17.1: Default defense time allocation

Stage	Minutes	Notes
Presentation	20	Every member presents part of the work.
Demonstration	10	Live on the group's own prepared environment.
Questions	20	Group and individual questions.
Panel deliberation	10	Without the group present.

The demonstration should be live. A recorded demonstration may replace it only when the Committee has approved that in advance, for example for specialized hardware, and the group should always have a recorded backup. The Coordinator communicates the outcome in writing within two working days.

17.3 Presentation and demonstration protocols

The defense also assesses professional conduct. Panels consider the following.

- Dress should be professional and consistent with the University code of conduct. Confidence, the quality of the slides and appropriate use of time are also considered.
- Presenting well is one of the skills a student is expected to develop. Examiners judge whether the student answers the question asked, defends the work accurately, and acknowledges weaknesses and limitations honestly.
- In the demonstration, the group must map the implementation to the problem statement, to the functional and non-functional requirements it promised, and to the design. The implementation must be traceable to the documents the group produced.
- Source code is checked for originality and readability, for example the use of comments and adherence to good coding practice.

17.4 Assessment principle

Scoring must be evidence-based. Every mark must rest on identifiable evidence in the documents, the system, the logbook, the repository or the defense, and each examiner records the justification for each criterion on the assessment sheet (Appendix N). A polished presentation cannot make up for missing evidence.

17.5 Assessment components and weights

Each course mark combines the advisor's progress assessment (A) and the examiner assessment (E), as shown in Table 17.2. Both are scored from 0 to 100.

Table 17.2: Assessment components

Component	Final Year Project I	Final Year Project II
Advisor progress assessment A (Appendix C)	40%	40%
Examiner assessment E (rubric in Section 17.6)	60%	60%

The final mark for each student is calculated as follows:

$$Final\ mark = (0.40\ X\ A) + [0.60\ X\ (E\ X\ ICF)]$$

where:

- A is the student's individual advisor progress assessment, scored from 0 to 100 using Appendix C.
- E is the examiner assessment of the group's work, scored from 0 to 100 using the evaluation rubric in Table 17.3 (Section 17.6).
- ICF is the student's Individual Contribution Factor, assigned according to Section 17.7.

The ICF is applied only to the examiner assessment, E; it does not apply to the advisor assessment, A. Although students in the same group normally share the same examiner assessment, E, their final marks may differ because their individual advisor assessments and ICF values may differ.

17.6 Detailed 100-point evaluation rubric

Examiners use the rubric in Table 17.3 to score E. The advisor uses the form in Appendix C to score A. Weights differ between the two courses because implementation, testing and results are not yet available in Final Year Project I. An examiner first chooses the band for each criterion and then awards a mark inside that band in proportion to the criterion weight. The bands are Excellent at 90% to 100% of the weight, Good at 70% to 89%, Adequate at 50% to 69%, Weak at 30% to 49% and Poor at 0% to 29%.

Table 17.3: Evaluation rubric: weights and performance anchors

Criterion	Final Year Project I	Final Year Project II	Performance anchors
Problem definition and significance	15	10	**Excellent**: highly relevant, precise and significant, supported by evidence from the group or from credible sources, with stakeholders and gap clearly identified.

			Good: clear and relevant, with some evidence and a stated gap. **Adequate**: understandable, but the evidence or the justification of significance is weak. **Weak**: vague or too broad, with little or no evidence. **Poor**: poorly defined or not stated.
Literature review and related work	15	10	**Excellent**: comprehensive, current, organized by theme, critically synthesized, and leading to a clearly justified gap. **Good**: relevant and well organized, with some critical comparison and a stated gap. **Adequate**: mostly descriptive, with sources summarized one by one and little comparison. **Weak**: few sources, little organization, or a gap that is not supported. **Poor**: minimal or inappropriate sources and no identifiable gap.
Objectives and requirements	20	10	**Excellent**: measurable, aligned with the problem, complete, and traceable to design and tests. **Good**: clear and mostly measurable, with minor gaps in completeness or traceability. **Adequate**: some objectives or requirements are vague or not measurable.

			Weak: inconsistent or incomplete, with weak links between objectives and requirements. **Poor**: absent or inappropriate.
Methodology	15	10	**Excellent**: appropriate, justified against alternatives, and rigorously applied to every specific objective. **Good**: appropriate and reasonably justified, and applied to most objectives. **Adequate**: identified but weakly justified or inconsistently applied. **Weak**: poorly suited to the objectives or described without detail. **Poor**: absent.
Analysis and design	25	15	**Excellent**: technically correct, complete, consistent across models, and traceable to requirements. **Good**: sound, with minor gaps or inconsistencies. **Adequate**: several gaps or inconsistencies, but the main design is usable. **Weak**: incomplete or contradictory models with little link to requirements. **Poor**: little or no design evidence.
Technical implementation	0	20	**Excellent**: robust, fully functional, maintainable and secure, with the technical challenge clearly met and the work reproducible from the submission.

			Good: functional and well structured, with minor weaknesses. **Adequate**: functional, with noticeable limitations in scope, quality or security. **Weak**: partial implementation, or major parts not working. **Poor**: non-functional or very limited.
Testing and evaluation	0	10	**Excellent**: systematic, covers all requirements, test levels justified by the test plan, and evidence-rich. **Good**: covers most requirements with adequate evidence. **Adequate**: basic tests with limited evidence. **Weak**: few tests and no link to requirements. **Poor**: absent or unverifiable.
Results, analysis and discussion	0	5	**Excellent**: accurate and critical, with every objective linked to evidence and an achievement status, and with limitations and error analysis included. **Good**: accurate and linked to most objectives, with some critical discussion. **Adequate**: results are reported, but the discussion is descriptive and links to objectives are limited. **Weak**: results are incomplete, unclear or only loosely linked to objectives. **Poor**: absent or unreliable.

Documentation and report quality	5	5	**Excellent**: complete, well organized, following the required structure and format, in clear technical English with correct IEEE citations. **Good**: complete and clear, with minor errors in format, language or citation. **Adequate**: mostly complete, with noticeable gaps or errors in structure, format, language or citation. **Weak**: incomplete or poorly organized, with frequent errors. **Poor**: unacceptable structure, language or referencing.
Presentation, demonstration and defense	5	5	**Excellent**: clear, well timed and confident, with accurate answers to technical questions and a demonstration mapped to the requirements. **Good**: clear and accurate, with minor weaknesses in timing, delivery or answers. **Adequate**: understandable, with some unclear or inaccurate answers or a limited demonstration. **Weak**: unclear delivery, poor timing, or answers that show limited understanding. **Poor**: unable to demonstrate or defend the work.
Total	**100**	**100**	

17.7 Individual Contribution Factor

The examiners, with the advisor's input, assign each student an Individual Contribution Factor (ICF) based on defense questioning, the logbook, the repository history and Appendix G. The ICF applies only to the examiner assessment, E. An ICF of 1.00 leaves E unchanged, while an ICF below 1.00 reduces the examiner component of the student's mark. The ICF must not increase E above the original group examiner assessment. The ICF does not impose a ceiling on the final individual mark; the final mark is calculated using Section 17.5 and also includes the student's individual advisor assessment, A. Table 17.4 gives the ICF anchors.

Table 17.4: Individual Contribution Factor anchors

ICF	Evidence
1.00	Explains the whole system and their own modules in depth, answers troubleshooting questions without the report, and the records show an equal or larger share of the work.
0.90	Explains the system and own modules well, with minor gaps.
0.80	Explains own modules but the understanding of other parts is limited.
0.70	Explains own modules only partly, or the records show a clearly smaller share of work.
0.60	Struggles to explain own modules, and the contribution is minimal.
Below 0.60	Cannot explain substantial submitted work. The examiners refer the case to the Committee under Section 18.1.

17.8 Moderation and final mark

Each examiner scores independently with Appendix N before the deliberation. Where the panel has one examiner, the total given by that examiner is the examiner mark E. Where the panel has two or more examiners, E is the average of their totals. If the highest and the lowest totals differ by more than 15 marks, the Committee appoints an additional examiner or reviews the evidence itself, and it decides the mark. The Coordinator consolidates the marks, and they stay provisional until the Committee approves them. Letter grades follow the University's senate legislation. Unless that legislation sets a different threshold, a student passes the project course

when the final mark reaches the University minimum pass mark and E is at least 50.

17.9 Defense outcomes

The panel records one of the following outcomes and gives the group a written list of required corrections within two working days.

- Pass: no corrections needed.
- Pass with minor corrections: the group completes them within two weeks and the advisor verifies them.
- Pass with major corrections: the group completes them within four weeks, and an examiner verifies them or re-examines the affected parts.
- Fail: the case follows Section 18.3.

The final mark is released after the corrections are verified.

18. Critical Failure, Re-submission and Appeals

18.1 Critical-failure conditions

The following conditions may require failure, re-submission or disciplinary action under university regulations:

- serious academic misconduct;
- fabricated results or data;
- inability to explain substantial submitted work, including AI-assisted work;
- no meaningful implementation where implementation is an approved objective;
- absence of meaningful testing or evaluation;
- major ethical or security violations;
- failure to demonstrate the claimed solution; or
- a project that differs materially from the approved proposal without an approved change.

18.2 Procedure

An examiner or advisor who identifies such a condition reports it in writing to the Committee with the supporting evidence. The Committee informs the students concerned and allows them to respond in writing or in person within three working days. The Committee records its decision and withholds the affected marks until then. Cases of academic misconduct are referred to the University disciplinary process.

18.3 Re-submission and re-defense

For a failed project the Committee issues a corrective plan that lists the deficiencies, the new deadline and the re-defense date. The re-defense takes place in the next available session. The number of attempts and any cap on the re-submission mark follow University legislation.

18.4 Appeals

A student may appeal in writing to the Department Head within five working days after the result is announced. The appeal must state its grounds, such as a procedural irregularity, a calculation error or evidence of bias, and disagreement with the academic judgment alone is not a ground. A panel that does not include the original examiners reviews the appeal and gives its decision within ten working days. After that the University's appeal procedures apply.

18.5 Special circumstances

Students with documented illness, disability, bereavement or other serious circumstances may request reasonable adjustments, such as extra time or a different format. They must apply in writing, with documentation, before the relevant deadline or, where this was not possible, within three days after it, as in Section 16.3. The Committee handles these requests in confidence and in line with university policy.

19. Quality Assurance and Continuous Improvement

19.1 Annual review

The FYP Committee or Coordinators should review each year the common failure patterns, the reliability of the rubric, project complexity, supervision workload, defense feedback and graduate outcomes.

19.2 Evidence for the review

The review should use the distribution of marks by examiner, similarity results, appeals and complaints, advisor load, surveys of students, advisors and examiners, feedback from alumni and employers, and project outcomes such as deployment, publication or later use by the sponsor.

19.3 Revision and version control

Record each revision of this guideline in the Document Control page that follows the title page, and approve it through the appropriate academic governance process. Update the templates at the same time and announce the revised guideline before the semester starts.

Appendix A: Project Proposal Template

Complete the following. Use the section order below.

Title	
Student(s) and ID(s)	
Advisor	
Co-advisor or sponsor (if any)	
Primary project category (Section 2.3)	
Proposed complexity class (Section 2.4)	

1. Background
2. Problem statement and gap
3. Motivation
4. Objectives (general objective and specific objectives, without section numbers)
5. Research or development questions (where applicable)
6. Scope and limitations
7. Significance
8. Literature review and related work (with comparison table)
9. Methodology
10. Preliminary requirements
11. Proposed design and technologies
12. Resources, risks and ethical considerations (attach Appendix M)
13. Expected deliverables
14. Work plan

References

Approval block

Role	Decision	Signature	Date
Advisor (recommendation)	Recommend / Do not recommend		
Coordinator / Committee	Approved / Approved with modifications / Rejected		

Appendix B: Weekly Project Progress Log

Week and date	Member(s)	Objective addressed	Planned activity	Completed activity	Problems and risks	Advisor feedback	Agreed actions and next milestone	Advisor initials

Appendix C: Advisor Progress Assessment

Complete one form for each student. This assessment determines readiness for defense and gives the advisor component A in Section 17.5.

Student name and ID	
Project title	

Criterion	Weight	Score	Comments
Regularity and preparedness	10%		
Progress against milestones	20%		
Technical development	20%		
Problem solving and initiative	15%		
Documentation and logbook	10%		
Response to feedback	10%		
Professional conduct	10%		
Individual contribution	5%		
Total	**100%**		

Advisor recommendation: ☐ Ready for defense ☐ Ready with minor corrections ☐ Not ready (reasons attached in writing)

Advisor name: _____

Signature: _____ Date: _____

Appendix D: Defense Checklist

Examiners use this list to confirm that each student has shown the following. Items marked (FYP II only) apply only at the Final Year Project II defense. At the Final Year Project I defense, examiners assess the remaining items on the basis of the Final Year Project I document and presentation.

☐ The student can state the problem and justify its significance.

☐ The student can explain every specific objective and its evidence.

☐ The student can explain the methodology and justify the major choices.

☐ The student can explain the system architecture and design.

☐ The student can demonstrate the major implemented functionality. (FYP II only)

☐ The student can explain the database, data or model design.

☐ The student can explain the testing and the failures encountered. (FYP II only)

☐ The student can interpret the evaluation results. (FYP II only)

☐ The student can state limitations honestly.

☐ The student can explain their individual contribution.

☐ The student can map the implementation to the requirements and design. (FYP II only)

☐ The student can explain any use of AI tools and the checks made on AI-assisted work.

☐ The student can discuss the ethical and security considerations of the project.

☐ The student can answer technical questions without relying on the report.

☐ The student can identify future improvements.

Appendix E: Objective-to-Evidence Traceability Matrix

Objective	Requirement(s)	Design artifact	Implementation or research output	Test/evaluation	Result	Achievement
Example (illustrative): SO2, implement a triage classifier	FR-03, NFR-02	Model design, Figure 4.5	Module classify.py	Test T-12, evaluation E-3	Recall of 0.87 on test set	Achieved

Achievement values: achieved, partially achieved, not achieved. Each claim must be supported by identifiable evidence.

Appendix F: Test Case Template

Test ID	Requirement	Precondition or input	Expected result	Actual result	Status	Evidence or defect ID	Severity if failed

Appendix G: Individual Contribution Statement

Student name and ID	
Group project title	

Area	Specific contribution	Evidence	Estimated effort (% of own time)
Requirements			
Design			
Implementation			
Database or data			
Testing			
Documentation			
Other			

Student declaration: I certify that the contribution described above accurately represents my work, subject to the University's academic-integrity rules.

Student signature: _____ Date: _____

Advisor acknowledgement: _____ Date: _____

Appendix H: Project Risk Register

Risk ID	Risk	Probability (L/M/H)	Impact (L/M/H)	Mitigation	Owner	Status
R1	Data unavailable					
R2	Technology failure					
R3	Scope expansion					
R4	Schedule delay					
R5	Security or privacy issue					
R6	Other					

Appendix I: Final Submission Checklist

☐ Approved final report (PDF and editable source)

☐ Source code and repository

☐ Database and scripts where applicable

☐ Executable or deployed system where applicable

☐ Dataset and data documentation, where permitted

☐ Installation guide

☐ User guide

☐ Testing evidence

☐ Evaluation results

☐ Presentation slides

☐ Project logbook

☐ Individual contribution statement

☐ Generative-AI use declaration (Appendix L)

☐ Similarity report

☐ Reproducibility package, for AI and data projects

☐ List of third-party libraries, data and licenses

☐ Ethical approval documents where applicable

☐ Required declarations and approvals

Appendix J: Technical Documentation Checklist

- System overview and architecture
- Installation prerequisites and dependencies
- Configuration and environment variables
- Database setup and migration instructions
- API documentation where applicable
- Authentication and authorization model
- Testing instructions
- Known limitations and defects
- Deployment instructions
- Backup and recovery considerations
- Third-party libraries and licenses
- Data description and model description for AI projects
- Security considerations

Appendix K: Title Page and Front-Matter Templates

Title page. Center all lines. Use Times New Roman and place the elements in this order:

1. University logo.
2. Addis Ababa University (16 pt, bold).
3. College of Natural and Computational Sciences (14 pt, bold).
4. Department of Computer Science (14 pt, bold).
5. Project title (16 pt, bold, upper case).
6. The statement: "A [Final Year Project I document / Final Year Project II report] submitted to the Department of Computer Science in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science" (12 pt).
7. Names and IDs of the group members, and the advisor's name (12 pt).
8. Place, month and year (12 pt).

Declaration. "We declare that this document is our original work, that it has not been submitted elsewhere for any award, and that all sources of information have been acknowledged. We have disclosed our use of AI tools in Appendix L."

Name and ID	Signature	Date

Approval sheet. “This document is submitted for examination with the approval of the university advisor.”

Advisor’s Name	Signature	Date

The examiners' names and signatures follow on the same sheet after the defense.

Appendix L: Generative-AI Use Disclosure

Project title	
Group members	

☐ We did not use generative-AI tools in this project.

☐ We used generative-AI tools as listed below.

Tool and version	Dates	Purpose	Where used (section or module)	How the output was checked

We confirm that all AI-assisted text and code was reviewed, understood and tested by us, that all references were verified, and that no confidential or personal data was entered into public AI services.

Signatures: _____ _____ _____

Date: _____

Appendix M: Ethical Screening Checklist

Answer each question. For every "Yes", describe the safeguards. A "Yes" to any of the first five questions requires the written approval described in Section 15.4.

Question	Yes / No	Safeguards and notes
Does the project involve human participants (surveys, interviews, user tests)?		
Does it collect or process identifiable personal data?		
Does it use health, biometric, financial or education records?		
Does it involve minors or other vulnerable people?		
Does it use confidential data from a sponsor?		
Does it collect data from websites or third-party platforms?		
Does it involve security testing or offensive techniques?		
Does it involve hardware with a safety risk?		
Will project data be sent to an external AI service?		
Could the system cause harm or unfair treatment if it fails or is biased?		

Decision by	Outcome	Signature	Date
Advisor			
Committee			

Appendix N: Examiner Assessment Sheet

Final Year Project (I or II)		
Group Members	1	
	2	
	3	
	4	

Criterion	Weight	Band	Mark	Evidence-based justification
Problem definition and significance				
Literature review and related work				
Objectives and requirements				
Methodology				
Analysis and design				
Technical implementation				
Testing and evaluation				
Results, analysis and discussion				
Documentation and report quality				
Presentation, demonstration and defense				
Total (E)	**100**			

Student	ICF	Evidence for ICF

Examiner name: _____ Signature: _____

Date: _____